1.59 Equations of Planes - Vector Equations

Worked Solutions

1 a)

Let A, B and C be $(1,0,1)$, $(2,1,0)$ and $(1,2,2)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}.$$

$$\therefore \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}.$$

1 b)

Let A, B and C be $(-1,2,-3)$, $(3,2,3)$ and $(-1,-1,2)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 6 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ 5 \end{pmatrix}.$$

$$\therefore \mathbf{r} = \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 0 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ -3 \\ 5 \end{pmatrix}.$$

1 c)

Let A, B and C be $(0,0,0)$, $(3,4,1)$ and $(-9,-3,7)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} -9 \\ -3 \\ 7 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -9 \\ -3 \\ 7 \end{pmatrix}.$$

$$\therefore \mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -9 \\ -3 \\ 7 \end{pmatrix} \Rightarrow \mathbf{r} = \lambda \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -9 \\ -3 \\ 7 \end{pmatrix}.$$

2 a)

$$\begin{pmatrix} 9 \\ 18 \\ -7 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

$$9 = 3 - \lambda + 2\mu \Rightarrow -\lambda + 2\mu = 6 \quad (1).$$

$$18 = 2 + 2\lambda + 3\mu \Rightarrow 2\lambda + 3\mu = 16 \quad (2).$$

$\lambda = 2$ and $\mu = 4$. Check:

$$-5 + 2(1) + 4(-1) = -7.$$

So $(9, 18, -7)$ lies on the plane.

2 b)

$$\begin{pmatrix} 24 \\ 23 \\ -17 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

$$24 = 3 - \lambda + 2\mu \Rightarrow -\lambda + 2\mu = 21 \quad (1).$$

$$23 = 2 + 2\lambda + 3\mu \Rightarrow 2\lambda + 3\mu = 21 \quad (2).$$

$\lambda = -3$ and $\mu = 9$. Check:

$$-5 - 3(1) + 9(-1) = -17.$$

So $(24, 23, -17)$ lies on the plane.

2 c)

$$\begin{pmatrix} 19 \\ 5 \\ -22 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

$$19 = 3 - \lambda + 2\mu \Rightarrow -\lambda + 2\mu = 16 \quad (1).$$

$$5 = 2 + 2\lambda + 3\mu \Rightarrow 2\lambda + 3\mu = 3 \quad (2).$$

$\lambda = -6$ and $\mu = 5$. Check:

$$-5 - 6(1) + 5(-1) = -16 \neq -22.$$

So $(19, 5, -22)$ does not lie on the plane.

3 a)

$$\overrightarrow{OC} = \lambda\overrightarrow{OA} + \mu\overrightarrow{OB} \Rightarrow \begin{pmatrix} 34 \\ -23 \\ 13 \end{pmatrix} = \lambda \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -6 \\ 2 \\ 3 \end{pmatrix}.$$

$$34 = 2\lambda - 6\mu \quad (1).$$

$$-23 = -3\lambda + 2\mu \quad (2).$$

$$\therefore \lambda = 5 \text{ and } \mu = -4.$$

Check:

$$5(5) - 4(3) = 13.$$

So $\overrightarrow{OC} = 5\overrightarrow{OA} - 4\overrightarrow{OB}$.

3 b)

This tells us that O, A, B and C are coplanar.

4 a)

$$\begin{pmatrix} 3 \\ -9 \\ 8 \end{pmatrix} \cdot \begin{pmatrix} 35 \\ 33 \\ 24 \end{pmatrix} = 3 \times 35 + (-9) \times 33 + 8 \times 24 = 105 - 297 + 192 = 0.$$

$$\begin{pmatrix} 3 \\ 7 \\ -14 \end{pmatrix} \cdot \begin{pmatrix} 35 \\ 33 \\ 24 \end{pmatrix} = 3 \times 35 + 7 \times 33 + (-14) \times 24 = 105 + 231 - 336 = 0.$$

$$\therefore 35\mathbf{i} + 33\mathbf{j} + 24\mathbf{k} \text{ is a normal vector to the plane.}$$

4 b)

The plane has equation $35x + 33y + 24z = d$.

$$d = 35(-2) + 33(1) + 24(4) = 59.$$

$$\therefore 35x + 33y + 24z = 59.$$

5)

First determine whether the point $(2, 0, 0)$ lies on the plane:

$$\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}.$$

$$2 = \alpha + \beta \Rightarrow \alpha + \beta = 2 \quad (1).$$

$$0 = 1 - 2\beta \Rightarrow \beta = \frac{1}{2} \quad (2).$$

Solve (1) and (2) to get $\alpha = \frac{3}{2}$ and $\beta = \frac{1}{2}$. Check:

$$0 - \left(\frac{3}{2}\right) + 3\left(\frac{1}{2}\right) = 0 \quad \text{as required.}$$

So $(2, 0, 0)$ lies on the plane. We also have that

$$\begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix} = - \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$$

so the direction vector of the line is parallel to one of the direction vectors of the plane. So the line lies in the plane.

6 a)

$$\mathbf{n}_1 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 4 \\ 3 & 1 & 0 \end{vmatrix} = (1 \times 0 - 4 \times 1)\mathbf{i} - (2 \times 0 - 4 \times 3)\mathbf{j} + (2 \times 1 - 1 \times 3)\mathbf{k}.$$

$$= -4\mathbf{i} + 12\mathbf{j} - \mathbf{k}.$$

$$\mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & -2 \\ 2 & 1 & -1 \end{vmatrix} = (3 \times (-1) - (-2) \times 1)\mathbf{i} - (1 \times (-1) - (-2) \times 2)\mathbf{j} + (1 \times 1 - 3 \times 2)\mathbf{k}.$$

$$= -\mathbf{i} - 3\mathbf{j} - 5\mathbf{k}.$$

$$\cos\theta = \frac{\begin{pmatrix} -4 \\ 12 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -3 \\ -5 \end{pmatrix}}{\sqrt{(-4)^2 + 12^2 + (-1)^2} \times \sqrt{(-1)^2 + (-3)^2 + (-5)^2}}.$$

$$\Rightarrow \cos\theta = \frac{(-4) \times (-1) + 12 \times (-3) + (-1) \times (-5)}{\sqrt{161} \times \sqrt{35}}.$$

$$\Rightarrow \cos\theta = -\frac{27\sqrt{115}}{805} \Rightarrow \theta = 111.080\ldots^\circ.$$

Acute Angle $= 68.9^\circ$ (1 DP).

6 b)

$$\mathbf{n}_1 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & 0 & 2 \\ 3 & -2 & 0 \end{vmatrix} = (0 \times 0 - 2 \times (-2))\mathbf{i} - (5 \times 0 - 2 \times 3)\mathbf{j} + (5 \times (-2) - 0 \times 3)\mathbf{k}.$$

$$= 4\mathbf{i} + 6\mathbf{j} - 10\mathbf{k}.$$

$$\mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 1 & 1 \\ 3 & -2 & 3 \end{vmatrix} = (1 \times 3 - 1 \times (-2))\mathbf{i} - ((-2) \times 3 - 1 \times 3)\mathbf{j} + ((-2) \times (-2) - 1 \times 3)\mathbf{k}.$$

$$= 5\mathbf{i} + 9\mathbf{j} + \mathbf{k}.$$

$$\cos\theta = \frac{\begin{pmatrix} 4 \\ 6 \\ -10 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 9 \\ 1 \end{pmatrix}}{\sqrt{4^2 + 6^2 + (-10)^2} \times \sqrt{5^2 + 9^2 + 1^2}}.$$

$$\Rightarrow \cos\theta = \frac{4 \times 5 + 6 \times 9 + (-10) \times 1}{\sqrt{152} \times \sqrt{107}}.$$

$$\Rightarrow \cos\theta = \frac{16\sqrt{4066}}{2033} \Rightarrow \theta = 59.878\ldots^\circ.$$

Acute Angle $= 59.9^\circ$ (1 DP).

7 a)

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix}.$$

$$= (0 \times 1 - 0 \times 1)\mathbf{i} - (1 \times 1 - 0 \times 0)\mathbf{j} + (1 \times 1 - 0 \times 0)\mathbf{k}.$$

$$= -\mathbf{j} + \mathbf{k}.$$

The plane has equation $-y + z = d$.

$$d = -(1) + (0) = -1.$$

$$-y + z = -1.$$

$$\Rightarrow y - z = 1.$$

7 b)

$$\begin{pmatrix}2\\1\\1\end{pmatrix} \times \begin{pmatrix}4\\3\\1\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\2 & 1 & 1\\4 & 3 & 1\end{vmatrix}.$$

$$= (1 \times 1 - 1 \times 3)\mathbf{i} - (2 \times 1 - 1 \times 4)\mathbf{j} + (2 \times 3 - 1 \times 4)\mathbf{k}.$$

$$= -2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}.$$

The plane has equation $-2x + 2y + 2z = d$.

$$d = -2(1) + 2(2) + 2(5) = 12.$$

$$-2x + 2y + 2z = 12.$$

$$\Rightarrow -x + y + z = 6.$$

7 c)

$$\begin{pmatrix}3\\2\\0\end{pmatrix} \times \begin{pmatrix}0\\-5\\6\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\3 & 2 & 0\\0 & -5 & 6\end{vmatrix}.$$

$$= (2 \times 6 - 0 \times (-5))\mathbf{i} - (3 \times 6 - 0 \times 0)\mathbf{j} + (3 \times (-5) - 2 \times 0)\mathbf{k}.$$

$$= 12\mathbf{i} - 18\mathbf{j} - 15\mathbf{k}.$$

The plane has equation $12x - 18y - 15z = d$.

$$d = 12(4) - 18(2) - 15(-5) = 87.$$

$$12x - 18y - 15z = 87.$$

$$\Rightarrow 4x - 6y - 5z = 29.$$

7 d)

$$\begin{pmatrix}-2\\4\\-7\end{pmatrix} \times \begin{pmatrix}5\\0\\-4\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\-2 & 4 & -7\\5 & 0 & -4\end{vmatrix}.$$

$$= (4 \times (-4) - (-7) \times 0)\mathbf{i} - ((-2) \times (-4) - (-7) \times 5)\mathbf{j} + ((-2) \times 0 - 4 \times 5)\mathbf{k}.$$

$$= -16\mathbf{i} - 43\mathbf{j} - 20\mathbf{k}.$$

The plane has equation $-16x - 43y - 20z = d$.

$$d = -16(-2) - 43(-8) - 20(10) = 176.$$

$$-16x - 43y - 20z = 176.$$

8 a)

$$\begin{pmatrix}2\\1\\5\end{pmatrix} \times \begin{pmatrix}-3\\-2\\9\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\2 & 1 & 5\\-3 & -2 & 9\end{vmatrix}.$$

$$= (1 \times 9 - 5 \times (-2))\mathbf{i} - (2 \times 9 - 5 \times (-3))\mathbf{j} + (2 \times (-2) - 1 \times (-3))\mathbf{k}.$$

$$= 19\mathbf{i} - 33\mathbf{j} - \mathbf{k}.$$

The plane has equation $19x - 33y - z = d$.

$$d = 19(1) - 33(-10) - (-2) = 351.$$

$$19x - 33y - z = 351.$$

8 b)

$$\begin{pmatrix}2\\3\\-15\end{pmatrix} \times \begin{pmatrix}8\\0\\7\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\2 & 3 & -15\\8 & 0 & 7\end{vmatrix}.$$

$$= (3 \times 7 - (-15) \times 0)\mathbf{i} - (2 \times 7 - (-15) \times 8)\mathbf{j} + (2 \times 0 - 3 \times 8)\mathbf{k}.$$

$$= 21\mathbf{i} - 134\mathbf{j} - 24\mathbf{k}.$$

The plane has equation $21x - 134y - 24z = d$.

$$d = 21(0) - 134(0) - 24(0) = 0.$$

$$21x - 134y - 24z = 0.$$

9 a)

$$\begin{pmatrix}0\\1\\3\end{pmatrix} \times \begin{pmatrix}-2\\3\\2\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\0 & 1 & 3\\-2 & 3 & 2\end{vmatrix}.$$

$$= (1 \times 2 - 3 \times 3)\mathbf{i} - (0 \times 2 - 3 \times (-2))\mathbf{j} + (0 \times 3 - 1 \times (-2))\mathbf{k}.$$

$$= -7\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}.$$

The plane has equation $-7x - 6y + 2z = d$.

$$d = -7(2) - 6(1) + 2(1) = -18.$$

$$-7x - 6y + 2z = -18.$$

$$\Rightarrow \begin{pmatrix}x\\y\\z\end{pmatrix} \cdot \begin{pmatrix}-7\\-6\\2\end{pmatrix} = -18.$$

$$\Rightarrow \mathbf{r} \cdot \begin{pmatrix}-7\\-6\\2\end{pmatrix} = -18.$$

9 b)

$$\begin{pmatrix}20\\-8\\5\end{pmatrix}\times\begin{pmatrix}15\\8\\0\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\20&-8&5\\15&8&0\end{vmatrix}.$$

$$=((-8)\times 0-5\times 8)\mathbf{i}-(20\times 0-5\times 15)\mathbf{j}+(20\times 8-(-8)\times 15)\mathbf{k}.$$

$$=-40\mathbf{i}+75\mathbf{j}+280\mathbf{k}.$$

The plane has equation $-40x+75y+280z=d$.

$$d=-40(0)+75(-8)+280(16)=3880.$$

$$-40x+75y+280z=3880.$$

$$\Rightarrow -8x+15y+56z=776.$$

$$\Rightarrow \begin{pmatrix}x\\y\\z\end{pmatrix}\cdot\begin{pmatrix}-8\\15\\56\end{pmatrix}=776.$$

$$\Rightarrow \mathbf{r}\cdot\begin{pmatrix}-8\\15\\56\end{pmatrix}=776.$$

10 a)

$$\cos\theta=\frac{\begin{pmatrix}2\\-3\\8\end{pmatrix}\cdot\begin{pmatrix}-4\\1\\2\end{pmatrix}}{\sqrt{2^2+(-3)^2+8^2}\times\sqrt{(-4)^2+1^2+2^2}}.$$

$$\Rightarrow\cos\theta=\frac{2\times(-4)+(-3)\times 1+8\times 2}{\sqrt{77}\times\sqrt{21}}.$$

$$\Rightarrow\cos\theta=\frac{5}{\sqrt{1617}}\Rightarrow\theta=82.857\ldots^\circ.$$

Acute angle is 82.9° (1 DP).

10 b)

First write Π_3 in cartesian form:

$$\begin{pmatrix}2\\6\\-3\end{pmatrix}\times\begin{pmatrix}5\\4\\-1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&6&-3\\5&4&-1\end{vmatrix}.$$

$$=(6\times(-1)-(-3)\times 4)\mathbf{i}-(2\times(-1)-(-3)\times 5)\mathbf{j}+(2\times 4-6\times 5)\mathbf{k}.$$

$$=6\mathbf{i}-13\mathbf{j}-22\mathbf{k}.$$

The plane has equation $6x-13y-22z=d$.

$$d=6(3)-13(-5)-22(6)=-49.$$

$$6x-13y-22z=-49.$$

Now find the angle:

$$\cos\theta = \frac{\begin{pmatrix}2\\-3\\8\end{pmatrix}\cdot\begin{pmatrix}6\\-13\\-22\end{pmatrix}}{\sqrt{2^2+(-3)^2+8^2}\times\sqrt{6^2+(-13)^2+(-22)^2}}.$$

$$\Rightarrow \cos\theta = \frac{2\times 6+(-3)\times(-13)+8\times(-22)}{\sqrt{77}\times\sqrt{689}}.$$

$$\Rightarrow \cos\theta = -\frac{125}{\sqrt{53053}} \Rightarrow \theta = 122.867\ldots^\circ.$$

Acute angle is $180^\circ - 122.867\ldots^\circ = 57.1^\circ$ (1 DP).

11)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}-3\\1\\0\end{pmatrix} - \begin{pmatrix}0\\2\\6\end{pmatrix} = \begin{pmatrix}-3\\-1\\-6\end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix}3\\0\\-4\end{pmatrix} - \begin{pmatrix}0\\2\\6\end{pmatrix} = \begin{pmatrix}3\\-2\\-10\end{pmatrix}.$$

$$\overrightarrow{AB}\times\overrightarrow{AC} = \begin{pmatrix}-3\\-1\\-6\end{pmatrix}\times\begin{pmatrix}3\\-2\\-10\end{pmatrix} = \begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-3&-1&-6\\3&-2&-10\end{vmatrix}.$$

$$= ((-1)\times(-10)-(-6)\times(-2))\mathbf{i} - ((-3)\times(-10)-(-6)\times 3)\mathbf{j} + ((-3)\times(-2)-(-1)\times 3)\mathbf{k}.$$

$$= -2\mathbf{i} - 48\mathbf{j} + 9\mathbf{k}.$$

The plane has equation $-2x - 48y + 9z + d = 0$. Substitute in the point $A(0,2,6)$ to find d:

$$-2(0) - 48(2) + 9(6) + d = 0 \Rightarrow d = 42.$$

So the plane has equation $-2x - 48y + 9z + 42 = 0$ or $2x + 48y - 9z - 42 = 0$.

12)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}2\\3\\12\end{pmatrix} - \begin{pmatrix}4\\-5\\1\end{pmatrix} = \begin{pmatrix}-2\\8\\11\end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix}6\\-2\\10\end{pmatrix} - \begin{pmatrix}4\\-5\\1\end{pmatrix} = \begin{pmatrix}2\\3\\9\end{pmatrix}.$$

$$\overrightarrow{AB}\times\overrightarrow{AC} = \begin{pmatrix}-2\\8\\11\end{pmatrix}\times\begin{pmatrix}2\\3\\9\end{pmatrix} = \begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-2&8&11\\2&3&9\end{vmatrix}.$$

$$= (8\times 9 - 11\times 3)\mathbf{i} - ((-2)\times 9 - 11\times 2)\mathbf{j} + ((-2)\times 3 - 8\times 2)\mathbf{k}.$$

$$= 39\mathbf{i} + 40\mathbf{j} - 22\mathbf{k}.$$

The plane has equation

$$\left(\mathbf{r} - \begin{pmatrix}4\\-5\\1\end{pmatrix}\right)\cdot\begin{pmatrix}39\\40\\-22\end{pmatrix} = 0.$$